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Semiconductor device

Abstract

A semiconductor device, including a semiconductor element; a case having a frame portion, which has an inner periphery that surrounds a housing space for accommodating the semiconductor element; and a lid covering the housing space. The inner periphery of the frame portion has a stepped portion formed thereon, the stepped portion including a step supporting surface positioned at a level lower than a front surface of the frame portion and being approximately parallel to the front surface of the frame portion. The lid has a lateral surface surrounding the lid, and a front surface and a bottom surface approximately parallel to the step supporting surface. The lid has a reservoir formed in the front surface thereof and extending from the lateral surface, the reservoir having a reservoir surface positioned at a level lower than the front surface of the lid.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2021-202831, filed on Dec. 14, 2021, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The embodiments discussed herein relate to a semiconductor device.

2. Background of the Related Art

(3) Semiconductor devices include power devices and are used as power conversion devices. The power devices are insulated gate bipolar transistors (IGBTs) and power metal oxide semiconductor field effect transistors (MOSFETs). A semiconductor device includes semiconductor chips including power devices, a case that accommodates the semiconductor chips, and a lid that is

arranged at the opening of the case to cover the semiconductor chips. The lid is firmly fixed to the opening of the case using an adhesive material.

(4) Please see, for example, Japanese Laid-open Patent Publication No. 2003-051560.

(5) Here, the adhesive material fixing the lid to the case may protrude from the opening of the case. For example, if the case does not have an area large enough to apply the adhesive material, the adhesive material may protrude out of the area. Such protruding adhesive material may be a waste. In addition, if the protruding adhesive material adheres to the lid and case, it means defacing the semiconductor device and reducing its aesthetic value. In addition, the protruding adhesive material may make the surroundings of the semiconductor device dirty. In order to set an area for applying the adhesive material such as to prevent the adhesive material from protruding, the case needs to be enlarged, which leads to increasing the size of the semiconductor device.

SUMMARY OF THE INVENTION

(6) According to one aspect, there is provided a semiconductor device including: a semiconductor element; a case including a frame portion, which is of a frame shape having an opening in a plan view of the semiconductor device, the frame portion having a front surface, and an inner periphery that surrounds a housing space for accommodating the semiconductor element; and a lid having a flat plate shape, the lid covering an upper side of the housing space, wherein the inner periphery of the frame portion has a stepped portion formed thereon, the stepped portion including a step supporting surface positioned at a level lower than the front surface of the frame portion and being approximately parallel to the front surface of the frame portion, the lid has a front surface and a bottom surface, and a lateral surface surrounding the lid, the lateral surface being bonded to the stepped portion via an adhesive material, such that the front surface and the bottom surface of the lid are approximately parallel to the step supporting surface, and the lid has a reservoir formed in the front surface thereof and extending from the lateral surface, the reservoir having a reservoir surface substantially lower than the front surface of the lid.

(7) The object and advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the claims.

(8) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a semiconductor device according to a first embodiment;

(2) FIG. 2 is a plan view of a semiconductor device according to a second embodiment;

(3) FIG. 3 is a plan view of the inside of the semiconductor device according to the second embodiment;

(4) FIG. 4 is a sectional view of the semiconductor device according to the second embodiment;

(5) FIG. 5 is a plan view of an area where a lid and a case are attached to each other in the semiconductor device according to the second embodiment;

(6) FIG. 6 is a sectional view of the area where the lid and the case are attached to each other in the semiconductor device according to the second embodiment;

(7) FIG. 7 is a sectional view of an area where a lid and a case are attached to each other in a semiconductor device according to a modification example 2-1 of the second embodiment;

(8) FIG. 8 is a plan view of an area where a lid and a case are attached to each other in a semiconductor device according to a modification example 2-2 of the second embodiment;

(9) FIG. 9 is a plan view of a semiconductor device according to a third embodiment;

(10) FIG. 10 is a plan view of the inside of the semiconductor device according to the third embodiment; and

(11) FIG. 11 is a sectional view of the semiconductor device according to the third embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(12) Hereinafter, some embodiments will be described with reference to the accompanying drawings. In the following description, the terms “front surface” and “top surface” refer to surfaces facing the +Z direction in semiconductor devices **1**, **10**, and **10a** of FIGS. **1**, **2** and **9**. Similarly, the term “up” refers to the +Z direction in the semiconductor devices **1**, **10**, and **10a** of FIGS. **1**, **2** and **9**. The terms “rear surface” and “bottom surface” refer to surfaces facing the -Z direction in the semiconductor devices **1**, **10**, and **10a** of FIGS. **1**, **2** and **9**. Similarly, the term “down” refers to the -Z direction in the semiconductor devices **1**, **10**, and **10a** of FIGS. **1**, **2** and **9**. The term “side surface” refers to a surface connecting a “front surface” or “top surface” and a “rear surface” or “bottom surface” in the semiconductor devices **1**, **10**, and **10a** of FIGS. **1**, **2** and **9**. For example, a “side surface” is a surface facing one of the $\pm X$ directions and $\pm Y$ directions in the semiconductor devices **1**, **10**, and **10a** of FIGS. **1**, **2** and **9**. The same directionality applies to all drawings. The terms “front surface,” “top surface,” “up,” “rear surface,” “bottom surface,” “down,” and “side surface” are used for convenience to describe relative positional relationships, and do not limit the technical ideas of the embodiments. For example, the terms “up” and “down” are not always related to the vertical directions to the ground. That is, the “up” and “down” directions are not limited to the gravity direction. In addition, in the following description, the term “principal component” refers to a component contained at a volume ratio of 80 vol % or more. An expression “being approximately parallel” means that the angle formed by two objects falls within the range of 170° to 190°, inclusive. An expression “being approximately perpendicular” means that the angle formed by two objects falls within the range of 85° to 95°, inclusive.

First Embodiment

(13) A semiconductor device **1** according to a first embodiment will be described with reference to FIG. **1**. FIG. **1** illustrates a semiconductor device according to the first embodiment. In this connection, the middle part of FIG. **1** is a plan view of the semiconductor device **1** illustrated in the upper part of FIG. **1**. The upper part of FIG. **1** is a sectional view taken along the dash-dotted line Y1-Y1 of the semiconductor device **1** illustrated in the middle part of FIG. **1**. The lower part of FIG. **1** is a sectional view taken along the dash-dotted line Y2-Y2 of the semiconductor device **1** illustrated in the middle part of FIG. **1**.

(14) The semiconductor device **1** of FIG. **1** includes a case **2** and a lid **3** to accommodate semiconductor elements therein. The case **2** has a frame portion **2a**, which has a frame shape in plan view, and a bottom portion **2d**. The frame portion **2a** surrounds a housing space **2c** that is an opening to accommodate the semiconductor elements. The semiconductor elements are not illustrated here. In addition, the housing space **2c** is rectangular in plan view. Therefore, the frame portion **2a** has an opening inner periphery **2e** that surrounds the four sides of the housing space **2c**. In addition, the frame portion **2a** has a stepped portion **2b** formed along the opening inner periphery **2e** on the front surface side of the frame portion **2a**. The case **2** may accommodate any components of the semiconductor device **1**, as well as the semiconductor elements. For example, such components include an insulated circuit substrate on which the semiconductor elements are disposed, electronic parts, and a printed circuit substrate. The frame portion **2a** is rectangular in plan view, and the housing space **2c** and opening inner periphery **2e** are rectangular as well. The rectangular shape is assumed to be an oblong shape here, but may be a square shape.

(15) The stepped portion **2b** is continuously formed in a loop shape along the opening inner periphery **2e** of the frame portion **2a**. The stepped portion **2b** includes a step inner peripheral surface **2b1** and a step supporting surface **2b2**. The step inner peripheral surface **2b1** is formed at the long sides and short sides of the opening inner periphery **2e** so as to face the center of the frame portion **2a** in plan view. That is, the step inner peripheral surface **2b1** is formed in parallel to the Z-X plane at the opposite long sides of the opening inner periphery **2e** and is also formed in parallel to the Z-Y plane at the opposite short sides of the opening inner periphery **2e**. The height of the

step inner peripheral surface **2b1** depends on the thickness of the lid **3**. FIG. **1** illustrates the case where the step inner peripheral surface **2b1** has the same height as the lid **3**. The step supporting surface **2b2** is continuously formed in a loop shape along the opening inner periphery **2e** so as to be orthogonal to the step inner peripheral surface **2b1** and to face the +Z direction (in parallel to the X-Y plane). The step supporting surface **2b2** is approximately parallel to the front surface of the frame portion **2a**. That is, the step supporting surface **2b2** does not have any chamfered portion or recessed portion. The width (the length in the direction from the opening inner periphery **2e** toward the center in plan view) of the step supporting surface **2b2** is set so as to support the lid **3**, which will be described later. The size of the opening inner periphery **2e** is set so that, when the lid **3** is attached to the stepped portion **2b**, a gap **G** is formed between the step inner peripheral surface **2b1** and the lateral surface **3c2** of the lid **3**. In this connection, in the illustration of FIG. **1**, the size of the lid **3** relative to the case **2** is larger than the actual size, and accordingly the stepped portion **2b** is illustrated in large size relative to the case **2**.

(16) The bottom portion **2d** is formed on the rear surface side of the frame portion **2a**. Referring to FIG. **1**, the bottom portion **2d** is flush with the rear surface of the frame portion **2a**. The bottom portion **2d** closes the housing space **2c** from the bottom. FIG. **1** illustrates the case where the bottom portion **2d** is integrally formed with the frame portion **2a**. Alternatively, the bottom portion **2d** may be formed separately from the frame portion **2a**. The bottom portion **2d** may be a metal base substrate, for example.

(17) The lid **3** has a flat plate shape, and is rectangular in plan view. The rectangular shape of the lid **3** corresponds to the shape of the opening of the case **2**. The lid **3** has the lateral surface **3c2** that surrounds the four sides (opposite long sides and opposite short sides) thereof. This lid **3** is bonded to the stepped portion **2b** of the frame portion **2a** of the case **2** using an adhesive material **4**. Therefore, the housing space **2c** of the case **2** is surrounded by the lid **3**, frame portion **2a**, and bottom portion **2d**. In this connection, the lid **3** is attached to the stepped portion **2b** of the frame portion **2a** of the case **2**, as will be described later. At this time, the bottom surface **3b** of the lid **3** is entirely level. For example, the outer edge of the bottom surface **3b** of the lid **3** is supported by the step supporting surface **2b2** of the stepped portion **2b**. The outer edge of the bottom surface **3b** of this lid **3** and the rest of the bottom surface **3b** are level with each other. The bottom surface **3b** of the lid **3** does not have any chamfered portion or recessed portion.

(18) The lid **3** also has a reservoir **3c**. The reservoir **3c** is formed on the outer periphery of the front surface **3a** of the lid **3**. The reservoir **3c** extends in the direction from the lateral surface **3c2** of the front surface **3a** of the lid **3** toward the inner side of the lid **3** (the center of the lid **3**) in plan view and has an area that is positioned at a level lower than the front surface **3a** of the lid **3** at least on the side where the lateral surface **3c2** is positioned. The reservoir **3c** of FIG. **1** includes a reservoir surface **3c1** as such a low-level area. The reservoir surface **3c1** is positioned at a level lower than the front surface **3a** of the lid **3** and is parallel to the front surface **3a**. The reservoir **3c** has inner surfaces **3c3** and **3c4** that are parallel to the long sides of the lid **3** and that connect to the reservoir surface **3c1**. The inner surfaces **3c3** and **3c4** connect the reservoir surface **3c1** and the front surface **3a**. In this connection, as illustrated in the upper part of FIG. **1**, the surfaces connecting the reservoir surface **3c1** and the front surface **3a** of the lid **3** are perpendicular to the front surface **3a** of the lid **3** and the reservoir surface **3c1**. In addition, referring to FIG. **1**, such a reservoir **3c** is formed at the center of each of the long sides and short sides in the front surface **3a** of the lid **3**. Alternatively, the reservoir **3c** may be continuously formed in a loop shape along the outer periphery of the front surface **3a**.

(19) The lid **3** configured as above is attached to the stepped portion **2b** of the case **2**, and the adhesive material **4** is applied to the gap **G** between the lateral surface **3c2** of the lid **3** and the step inner peripheral surface **2b1** of the stepped portion **2b** of the case **2** and spreads in the gap **G**. At this time, the adhesive material **4** enters the reservoirs **3c** of the lid **3** and is accumulated therein. This prevents the adhesive material **4** from protruding from the gap **G**.

(20) As described above, the semiconductor device **1** includes the case **2** for accommodating semiconductor elements and the lid **3**. The case **2** has the frame portion **2a** that has a flame shape in plan view and has the opening inner periphery **2e** to surround the housing space **2c** for accommodating the semiconductor elements. The lid **3** has a flat plate shape and covers the housing space **2c**. On the opening inner periphery **2e** of the frame portion **2a**, the stepped portion **2b** is formed, which includes the step supporting surface **2b2** that is positioned at a level lower than the front surface of the frame portion **2a** and is approximately parallel to the front surface of the frame portion **2a**. The lid **3** has the lateral surface **3c2** that surrounds the lid **3** and that is bonded to the stepped portion **2b** using the adhesive material **4**, and the bottom surface **3b** that is approximately parallel to the step supporting surface **2b2**. The reservoirs **3c** each having an area positioned at a level lower than the front surface **3a** of the lid **3** are formed from the lateral surface **3c2** toward the inner side of the lid **3** in the front surface **3a** of the lid **3**.

(21) The adhesive material **4** is applied to the gap **G** between the lateral surface **3c2** of the lid **3** and the step inner peripheral surface **2b1** of the stepped portion **2b** of the case **2** and spreads in the gap **G**. At this time, the adhesive material **4** enters the reservoirs **3c** of the lid **3** and is accumulated therein. This enables preventing the adhesive material **4** from protruding from the gap **G**, without the need of increasing the size of the semiconductor device **1**, and also prevents a waste of the adhesive material **4**. Since the adhesive material **4** is prevented from protruding, the adhesive material **4** does not adhere to the lid **3** or case **2**, which prevents defacing of the semiconductor device **1**. In addition, the adhesive material **4** enters the reservoirs **3c** as well as the gap **G**. Therefore, the bonding area of the adhesive material **4** to the lid **3** increases, as compared with the case where the reservoirs **3c** are not formed. Therefore, the bonding force of the lid **3** to the case **2** using the adhesive material **4** is improved.

Second Embodiment

(22) In a second embodiment, a semiconductor device will be described with reference to FIGS. **2** to **4**, more concretely than the semiconductor device **1** of the first embodiment. FIG. **2** is a plan view of the semiconductor device according to the second embodiment, FIG. **3** is a plan view of the inside of the semiconductor device according to the second embodiment, and FIG. **4** is a sectional view of the semiconductor device according to the second embodiment. In this connection, FIG. **3** is a plan view of the semiconductor device **10** without a lid **75** and sealing material **95**. FIG. **4** is a sectional view taken along the dash-dotted line X-X of FIG. **2**. The position of the dash-dotted line X-X of FIG. **2** corresponds to that of the dash-dotted line X-X of FIG. **3**.

(23) The semiconductor device **10** includes a semiconductor unit **30**, a printed circuit substrate **40**, a case **50** for accommodating the semiconductor unit **30** and printed circuit substrate **40**, and a lid **75**. The semiconductor unit **30** includes a main circuit substrate **20**, and first and second semiconductor chips **31** and **32** mounted on the main circuit substrate **20**. In this connection, the semiconductor unit **30** includes six pairs of the first and second semiconductor chips **31** and **32**. The main circuit substrate **20** includes an insulating plate **21**, circuit patterns **22**, and a metal base substrate **23**. The main circuit substrate **20** and the printed circuit substrate **40** are disposed adjacent to each other in plan view.

(24) For example, as the insulating plate **21**, an organic insulating layer or a ceramic substrate may be used. The organic insulating layer is made of a combination of a resin with low thermal resistance and a material with high thermal conductivity. Examples of the former resin include an epoxy resin and a liquid crystal polymer insulating resin. Examples of the latter material include boron nitride, aluminum oxide, and silicon oxide.

(25) The ceramic substrate is made of ceramics with high thermal conductivity. For example, the ceramics are made from a material containing aluminum oxide, aluminum nitride, or silicon nitride as a principal component. In addition, the insulating plate **21** is rectangular in plan view. The thickness of the insulating plate **21** is in the range of 0.5 mm to 2.0 mm, inclusive.

(26) The circuit patterns **22** form a predetermined circuit. The circuit patterns **22** are formed on the

front surface of the insulating plate **21**, as illustrated in FIG. 3, for example. On one circuit pattern **22** formed on the left side of the front surface of the insulating plate **21**, three pairs of the first and second semiconductor chips **31** and **32** are mounted. On the right side of this circuit pattern **22**, three circuit patterns **22** are formed on the front surface of the insulating plate **21** so as to be aligned in the +X direction. One pair of the first and second semiconductor chips **31** and **32** is mounted on each of the three circuit patterns **22**. On one circuit pattern **22** formed at the lower right corner on the front surface of the insulating plate **21**, nothing is mounted.

(27) In addition, with respect to these circuit patterns **22**, their regions where the first and second semiconductor chips **31** and **32** are disposed are aligned in the upper portion of the insulating plate **21** as seen in FIG. 3, and the remaining regions extend downward on the insulating plate **21** as seen in FIG. 3. FIG. 3 merely illustrates the case where the total six pairs of the first and second semiconductor chips **31** and **32** are disposed on the circuit patterns **22**. The number of pairs of the first and second semiconductor chips **31** and **32** is not limited to six, but the number of pairs may be determined according to the specifications and others of the semiconductor device **10**, and the circuit patterns **22** may be provided according to the pairs. The plurality of circuit patterns **22** are formed on the front surface of the insulating plate **21**. The circuit patterns **22** are made of a metal with high electrical conductivity as a principal component. Examples of the metal include silver, copper, nickel, and an alloy containing at least one of these. The thicknesses of the circuit patterns **22** are preferably in the range of 0.1 mm to 1.0 mm, inclusive, and more preferably in the range of 0.2 mm to 0.5 mm. Plating may be performed on the surfaces of the circuit patterns **22** to improve their corrosion resistance. Examples of the plating material used here include nickel, a nickel-phosphorus alloy, and a nickel-boron alloy. The circuit patterns **22** are formed by etching a conductive plate or foil formed on one surface of the insulating plate **21** or by attaching conductive plates to one surface of the insulating plate **21**.

(28) The metal base substrate **23** is made of a metal with high thermal conductivity as a principal component. The corners of the metal base substrate **23** may be rounded or chamfered. Examples of the metal include aluminum, iron, silver, copper, and an alloy containing at least one of these. In addition, the metal base substrate **23** is rectangular in plan view and corresponds to a main circuit region **61** and control circuit region **62** (which will be described later) at the bottom portion **60** of the case **50**. The thickness of the metal base substrate **23** is in the range of 0.5 mm to 2.0 mm, inclusive. Plating may be performed on the surface of the metal base substrate **23** to improve its corrosion resistance. Examples of the plating material used here include nickel, a nickel-phosphorus alloy, and a nickel-boron alloy.

(29) In this connection, in the case where the insulating plate **21** is a ceramic substrate and the metal base substrate **23** is a metal foil, a direct copper bonding (DCB) substrate or an active metal brazed (AMB) substrate may be used as the circuit patterns **22**, insulating plate **21**, and metal base substrate **23**. In this connection, the shapes, positions, and quantity of the circuit patterns **22** and the positions and quantity of the first and second semiconductor chips **31** and **32** in the semiconductor unit **30** configured as above are just as an example and are not limited to those illustrated in FIGS. 2 to 4, but may be determined according to design and others.

(30) The first and second semiconductor chips **31** and **32** are power semiconductor chips made of silicon, silicon carbide, or gallium nitride. The first semiconductor chips **31** include a switching element. The switching element is an IGBT or a power MOSFET, for example. In the case where a first semiconductor chip **31** is an IGBT, the first semiconductor chip **31** has a collector electrode serving as a main electrode on the rear surface thereof and has a gate electrode serving as a control electrode and an emitter electrode serving as a main electrode on the front surface thereof. In the case where a first semiconductor chip **31** is a power MOSFET, the first semiconductor chip **31** has a drain electrode serving as a main electrode on the rear surface thereof and has a gate electrode serving as a control electrode and a source electrode serving as a main electrode on the front surface thereof. The rear surfaces of the first semiconductor chips **31** are bonded to the circuit

patterns **22** via a bonding material (not illustrated). As the bonding material, a solder or a sintered metal is used in the present embodiment. The solder is a lead-free solder containing a predetermined alloy as a principal component. For example, the predetermined alloy is any one of a tin-silver alloy, a tin-zinc alloy, and a tin-antimony alloy. The solder may contain an additive such as copper, bismuth, indium, nickel, germanium, cobalt, or silicon. As the sintered metal, aluminum or copper is used, for example.

(31) The second semiconductor chips **32** include a diode element. The diode element is a free wheeling diode (FWD) such as a Schottky barrier diode (SBD) or a P-intrinsic-N (PiN) diode, for example. A second semiconductor chip **32** of this type has an output electrode (a cathode electrode) serving as a main electrode on the rear surface thereof and has an input electrode (an anode electrode) serving as a main electrode on the front surface thereof. The rear surfaces of the second semiconductor chips **32** are bonded to the circuit patterns **22** using the bonding material.

(32) The thicknesses of these first and second semiconductor chips **31** and **32** are in the range of 180 μm and 220 μm , inclusive, for example, and are 200 μm on average. In place of the first and second semiconductor chips **31** and **32**, reverse-conducting (RC)-IGBT chips that have the functions of both IGBT and FWD may be used.

(33) The printed circuit substrate **40** is disposed adjacent to the main circuit substrate **20** disposed via an adhesive material **60a** in parallel to the bottom surface of the case **50**, and is positioned above the main circuit substrate **20** in the +Z direction. This printed circuit substrate **40** includes an insulating plate and a plurality of upper circuit patterns formed on the front surface of the insulating plate. In addition, the printed circuit substrate **40** may include a plurality of lower circuit patterns formed on the rear surface of the insulating plate.

(34) The insulating plate is made of an insulating material in a flat plate shape. As such a material, a material obtained by immersing a base in a resin is used. Examples of the base includes a paper, glass cloth, and glass unwoven fabric. Examples of the resin include a phenolic resin, an epoxy resin, and a polyimide resin. Specific examples of the insulating plate include a paper phenolic board, a paper epoxy board, a glass epoxy board, a glass polyimide board, and a glass composite board. The insulating plate is rectangular in plan view. The corners of the insulating plate may be rounded or chamfered.

(35) The plurality of upper circuit patterns and lower circuit patterns have predetermined different pattern shapes so as to implement a predetermined circuit. The upper circuit patterns and lower circuit patterns are made of a material with high electrical conductivity. Examples of this material include silver, copper, nickel, and an alloy containing at least one of these. Plating may be performed on the surfaces of the upper circuit patterns and lower circuit patterns to improve their corrosion resistance. Examples of the plating material used here include nickel, a nickel-phosphorus alloy, and a nickel-boron alloy.

(36) For example, the printed circuit substrate **40** is formed as follows. A metal foil is attached to each of the front surface and rear surface of the insulating plate, and resists in predetermined shapes are printed thereon. Then, the metal foils on the front surface and rear surface of the insulating plate are etched using the printed resists as masks, and the remaining resists are removed. By doing so, the upper circuit patterns and lower circuit patterns are obtained on the front surface and rear surface of the insulating plate.

(37) In addition, the printed circuit substrate **40** has control integrated circuits (ICs) **41** as electronic parts, which are electrically connected to the upper circuit patterns. In the present embodiment, the control ICs **41** are electrically and mechanically connected to the gate electrodes (control electrodes) of the first semiconductor chips **31** via bonding wires **43**, as illustrated in FIG. 3. The control ICs **41** apply control voltage to the first semiconductor chips **31** at predetermined timing. The bonding wires **43** used here are made of a material with high electrical conductivity. Examples of the material include gold, silver, copper, aluminum, and an alloy containing at least one of these. The diameters of the bonding wires **43** are in the range of 100 μm to 250 μm , inclusive, for

example. In this connection, other electronic parts may be mounted on the printed circuit substrate **40** as needed, in addition to the control ICs **41**. Examples of such electronic parts include a thermistor, a capacitor, a resistor, a current sensor, and a temperature sensor.

(38) In addition, in the main circuit substrate **20**, first and second semiconductor chips **31** and **32** are electrically and mechanically connected to each other with bonding wires **33**, and a second semiconductor chip **32** and a circuit pattern **22** are electrically and mechanically connected to each other with bonding wires **33**. In addition, the printed circuit substrate **40** and circuit patterns **22** of the main circuit substrate **20** are electrically and mechanically connected with bonding wires **33**. The bonding wires **33** used here are made of a material among the above-listed materials with high electrical conductivity. The diameters of the bonding wires **33** are in the range of 400 μm to 1.00 mm, inclusive, for example.

(39) The case **50** will now be described. The case **50** has the bottom portion **60** and a frame portion **70** that is integrally formed with the bottom portion **60** and that surrounds the periphery of the bottom portion **60**. In addition, the case **50** includes main current connection terminals **80a** to **80e** and control terminals **90**. In the following, the main current connection terminals **80a** to **80e** are collectively referred to as “main current connection terminals **80**” and individually as “main current connection terminal **80**” when the distinction among them is not needed.

(40) The case **50** is made of a thermoplastic resin. Examples of the thermoplastic resin includes a polyphenylene sulfide (PPS) resin, a polybutylene terephthalate (PBT) resin, a polybutylene succinate (PBS) resin, a polyamide (PA) resin, and an acrylonitrile-butadiene-styrene (ABS) resin. The case **50** is formed from such a thermoplastic resin by injection molding so as to be integrally formed with the main current connection terminals **80** and control terminals **90**.

(41) The bottom portion **60** is rectangular in plan view. On the bottom surface that is the front surface of the bottom portion **60**, the main circuit region **61** and control circuit region **62** are set. The main circuit region **61** of the bottom surface is an opening. The main circuit substrate **20** is disposed in the main circuit region **61**, and the printed circuit substrate **40** is disposed in the control circuit region **62**. That is, as illustrated in FIG. 3, the main circuit region **61** and control circuit region **62** are set adjacent to each other in plan view on the front surface of the bottom portion **60**. In this connection, the opening edge of the main circuit region **61**, which is an opening, on the rear surface side thereof may be tapered. In addition, a rear opening **63** is formed in the rear surface of the bottom portion **60**. In this connection, the area of the rear opening **63** corresponding to the main circuit region **61** penetrates through the bottom portion **60**. The semiconductor unit **30** is attached to the bottom portion **60** from the rear surface side. That is, the metal base substrate **23** is attached to the rear opening **63**, and the main circuit substrate **20** is exposed in the main circuit region **61** that is an opening. In this connection, the metal base substrate **23** is attached to the rear opening **63** of the bottom portion **60** using an adhesive material (not illustrated). The rear surface of thus attached metal base substrate **23** protrudes from the rear surface of the bottom portion **60** to the outside (downward in FIG. 4). Alternatively, the rear surface of the metal base substrate **23** may be flush with the rear surface of the bottom portion **60**.

(42) As the adhesive material for bonding the metal base substrate **23** to the rear opening **63** of the bottom portion **60**, a thermosetting resin-based adhesive agent or an organic-based adhesive agent is used, for example. For example, the thermosetting resin-based adhesive agent contains an epoxy resin or a phenolic resin as a principal component. For example, the organic-based adhesive agent is an elastomer-based adhesive agent containing silicone rubber or chloroprene rubber as a principal component. The adhesive material preferably contains an epoxy resin or silicone rubber as a principal component.

(43) On the front surface of the bottom portion **60**, the control circuit region **62** is set adjacent to the main circuit region **61** at the side thereof opposite to the side wall **71a**. The printed circuit substrate **40** is disposed in the control circuit region **62** of the bottom surface of the bottom portion **60** using the above-described bonding material.

(44) The frame portion **70** has a frame shape in plan view. The frame portion **70** has side walls **71a**, **71b**, **71c**, and **71d** that surround the four sides of the bottom portion **60**. In addition, the side walls **71a**, **71b**, **71c**, and **71d** respectively have opening inner peripheries that surround the housing space **72**. Out of the opening inner peripheries, FIG. **4** illustrates the opening inner peripheries **74a** and **74c** included in the side walls **71a** and **71c**. The opening inner peripheries included in the side walls **71b** and **71d** are provided in the same manner as the opening inner peripheries **74a** and **74c** although the illustration of them is omitted. The side walls **71a**, **71b**, **71c**, and **71d** all have the same height. The side walls **71a** and **71c** are provided at the long sides of the bottom portion **60**. The main current connection terminals **80a** to **80e** are formed by integral molding with the side wall **71a** and bottom portion **60** so as to be arranged along the side wall **71a**. The control terminals **90** are formed by integral molding with the side wall **71c** and bottom portion **60** so as to be arranged along the side wall **71c**.

(45) In addition, stepped portions **72a**, **72b**, **72c**, and **72d** are formed along the inner periphery on the front surface side of the side walls **71a**, **71b**, **71c**, and **71d**. In this connection, the stepped portions **72b** and **72d** respectively formed on the side walls **71b** and **71d** extend along the edges of fastening holes **73**, which will be described later. Projections **72a3**, **72b3**, **72c3**, and **72d3** are formed on the stepped portions **72a**, **72b**, **72c**, and **72d**, respectively. These stepped portions **72a**, **72b**, **72c**, and **72d** will be described in detail later.

(46) The main current connection terminal **80** has an L shape in side view, as illustrated in FIG. **4**. More specifically, the main current connection terminal **80** is formed by bending a plate member in an L shape. For example, The thickness of the main current connection terminal **80** is greater than or equal to 100 μm and less than 1.0 mm, and the width thereof is in the range of 1.0 mm to 10 mm, inclusive.

(47) The main current connection terminal **80a** has an external connection portion **81a** and an internal connection portion **82a**. One end portion of the external connection portion **81a** extends upward from the top surface of the side wall **71a**, and the other end portion thereof is integrally connected to one end of the internal connection portion **82a** inside the side wall **71a** and bottom portion **60**. The other end portion of the internal connection portion **82a** is exposed on a terminal stepped portion **71e** of the bottom portion **60**. The other end portion of the internal connection portion **82a** is electrically and mechanically connected to a circuit pattern **22** with the bonding wires **33**. Similarly, the main current connection terminals **80b** to **80e** have external connection portions **81b** to **81e** and internal connection portions **82b** to **82e**, respectively. One end portion of each of the external connection portions **81b** to **81e** extends upward from the top surface of the side wall **71a**, and the other end portion thereof is integrally connected to one end of the corresponding internal connection portion **82b** to **82e** inside the side wall **71a** and the bottom portion **60**. The other end portions of the internal connection portions **82b** to **82e** are exposed from the bottom portion **60**. The other end portions of the internal connection portions **82b** to **82e** are electrically and mechanically connected to the circuit patterns **22** with the bonding wires **33**. In this connection, as illustrated in FIG. **4**, the external connection portions **81a** to **81e** and internal connection portions **82a** to **82e** of the main current connection terminals **80a** to **80e** are collectively referred to as “external connection portions **81**” and “internal connection portions **82**” and individually as “external connection portion **81**” and “internal connection portion **82**” when the distinction among them is not needed. The main current connection terminal **80** is made of a material with high electrical conductivity. Examples of the material include copper, aluminum, nickel, and an alloy containing at least one of these. Plating may be performed on the surface of the main current connection terminal **80** to improve its corrosion resistance. Examples of the plating material used here include nickel and a nickel alloy.

(48) As illustrated in FIG. **4**, each control terminal **90** has an L shape in side view. More specifically, the control terminal **90** is formed by bending a columnar member in an L shape. The control terminal **90** has an external terminal portion **91** and an internal terminal portion **92**. One end

portion of the external terminal portion **91** extends upward from the top surface of the side wall **71c**, and the other end portion thereof is integrally connected to one end of the internal terminal portion **92** inside the side wall **71c** and the bottom portion **60**. The other end portion of the internal terminal portion **92** is exposed on a terminal stepped portion **71f** of the bottom portion **60**. The other end portion of the internal terminal portion **92** is electrically and mechanically connected to the printed circuit substrate **40** with a bonding wire **33**. In this manner, the internal terminal portion **92** of the control terminal **90** is electrically connected to the upper circuit pattern or lower circuit pattern of the printed circuit substrate **40**. When a control signal is input from the outside to the external terminal portion **91** of the control terminal **90**, the internal terminal portion **92** of the control terminal **90** and the printed circuit substrate **40** are made conductive. Then, when a control signal is input from the printed circuit substrate **40** to a control IC **41**, the control signal is output from the control IC **41** to the gate electrode of a first semiconductor chip **31** via a bonding wire **43**. (49) The control terminals **90** are made of a material with high electrical conductivity. Examples of the material include copper, aluminum, nickel, and an alloy containing at least one of these. Plating may be performed on the surface of the control terminals **90** to improve their corrosion resistance. Examples of the plating material used here include nickel and a nickel alloy.

(50) The side walls **71b** and **71d** are provided at the short sides of the bottom portion **60**, respectively. The fastening holes **73** for attaching a cooling unit to the rear surface of the semiconductor device **10** are provided in the side walls **71b** and **71d**, respectively. The cooling unit (not illustrated) is attached to the rear surface (rear surface of the metal base substrate **23**) of the semiconductor device **10** and is fixed thereto by inserting screws into the fastening holes **73**. At this time, the cooling unit may be attached to the rear surface of the semiconductor device **10** via a solder, a silver solder, a thermal grease, or a thermal sheet so as to improve the heat dissipation property. For example, the cooling unit in this case is made of a metal with high thermal conductivity as a principal component. Examples of the metal include aluminum, iron, silver, copper, and an alloy containing at least one of these. In addition, a heat sink or a cooling device using cool water may be used as the cooling unit, for example. In addition, the metal base substrate **23** may be integrally formed with such a cooling unit.

(51) A sealing material **95** is injected into the housing space **72** of the case **50** accommodating the above-described components, so that the housing space **72** is sealed with the sealing material **95**. The sealing material **95** contains a thermosetting resin and an inorganic filler, which is added to the thermosetting resin. For example, the thermosetting resin contains, as a principal component, at least one selected from the group of an epoxy resin, phenolic resin, and melamine resin. The thermosetting resin preferably contains the epoxy resin as a principal component. As the inorganic filler, an inorganic containing silicon oxide as a principal component is used. In addition, without the need of mixing flame retardant such as halogen-based, antimony-based, or metal hydroxide-based flame retardant, high flame retardancy is obtained. The content of the inorganic filler in the whole sealing material is in the range of 70 vol % to 90 vol %, inclusive.

(52) The lid **75** has a flat plate shape and is approximately rectangular in plan view. More specifically, the shape of the lid **75** in plan view corresponds to the shape of the opening of the housing space **72**. The lid **75** has lateral surfaces **75a2**, **75b2**, **75c2**, and **75d2** that surround the four sides thereof. In this connection, the lateral surfaces **75a2** and **75c2** correspond to the long sides of the lid **75** and are parallel to the $\pm X$ directions. The lateral surfaces **75b2** and **75d2** correspond to the short sides of the lid **75** and are parallel to the $\pm Y$ directions. In this connection, the centers of the lateral surfaces **75b2** and **75d2** are recessed toward the center due to the fastening holes **73** in plan view. In addition, when the lid **75** is attached to the stepped portions **72a**, **72b**, **72c**, and **72d** of the case **50**, as will be described later, the lateral surfaces **75a2**, **75b2**, **75c2**, and **75d2** are approximately parallel to the step inner peripheral surface (refer to a step inner peripheral surface **72c1** of FIG. 6). The thickness of the lid **75** is in the range of 1.8 mm to 2.2 mm, inclusive, for example, and may be set to 2.0 mm. Positioning members **76** and reservoirs **75c** are formed in the

front surface **75a** of the lid **75**. In addition, fixing holes (refer to a fixing hole **75b1** of FIG. **6**) are formed in the rear surface **75b** of the lid **75**. The fixing holes will be described in detail later.

(53) Two positioning members **76** are formed at each short side on the front surface **75a** of the lid **75**, for example. The positioning members **76** each have a columnar shape whose cross section decreases in the +Z direction, for example. A printed circuit substrate is separately attached to the positioning members **76**. The positions of the positioning members **76** are just as an example.

(54) A plurality of reservoirs **75c** are formed on the outer periphery of the opposite long sides in the front surface **75a** of the lid **75**. Here, four reservoirs **75c** are formed at each of the opposite long sides. The positions and quantity of the reservoirs **75c** are just as an example, and the reservoirs **75c** may be formed anywhere on the outer periphery of the lid **75**, and any number of reservoirs **75c** may be formed.

(55) In this connection, when the case **50** is fixed to a predetermined position via the fastening holes **73**, the case **50** may partly be curved and warped. If the case **50** is warped, the lid **75** may be separated from the frame portion **70** at the warped area even when the lid **75** is attached to the stepped portions **72a**, **72b**, **72c**, and **72d** using the adhesive material **96**. Therefore, the reservoirs **75c** may preferably be formed at positions of the outer periphery of the lid **75** corresponding to positions where the frame portion **70** is likely to be warped. The positions where the warpage is likely to occur depend on the positions of the fastening holes **73** in the case **50**. For example, in the present embodiment, the frame portion **70** is rectangular in plan view, and the fastening holes **73** are formed at the centers of the short sides of the frame portion **70**, respectively. In this case, the case **50** is probably warped in a convex shape in the +Z direction, along the center line passing through the center of each of the opposite long sides of the case **50**. Therefore, the reservoirs **75c** are preferably formed on the outer periphery of the long sides of the lid **75**.

(56) In addition, in the present embodiment, the positioning members **76** are formed at the short sides of the lid **75**. It is preferable not to form the reservoirs **75c** in the vicinity of the positioning members **76**. When a printed circuit substrate is attached to the positioning members **76**, stress (in the -Z direction) according to the attachment of the printed circuit substrate is generated in the vicinity of the connection areas of the positioning members **76** in the lid **75**. Note that the portions of the lid **75** where the reservoirs **75c** are formed are thinner than the rest of the lid **75** and are thus weak in strength. Therefore, if a reservoir **75c** is formed in the vicinity of such a connection area, the portion of the lid **75** where the reservoir **75c** is formed may be damaged in some cases. For this reason, the reservoirs **75c** are avoided from being formed at the short sides of the lid **75** where the positioning members **76** are provided. In this connection, the “vicinity” mentioned here may mean an area ranging from a positioning member **76** to a position apart from the positioning member **76** by a length corresponding to the depth of the stepped portion **72b** or **72d**.

(57) The lid **75** configured as above is firmly fixed to the stepped portions **72a**, **72b**, **72c**, and **72d** of the frame portion **70** using the adhesive material **96**. The fixed lid **75** covers the sealing material **95** filling the housing space **72**.

(58) The following describes an area where the lid **75** is attached to the case **50** (enclosed by a dashed line in FIGS. **2** and **4**) with reference to FIGS. **5** and **6**. FIG. **5** is a plan view of the area where the lid and the case are attached to each other in the semiconductor device according to the second embodiment. FIG. **6** is a sectional view of the area where the lid and the case are attached to each other in the semiconductor device according to the second embodiment. In this connection, FIG. **5** is an enlarged view of the area enclosed by the dashed line of FIG. **2**. FIG. **6** is a sectional view taken along the dash-dotted line X-X of FIG. **5**. In addition, a reservoir **75c** of the lid **75** attached to the stepped portion **72c** formed on the side wall **71c** will be described with reference to FIGS. **5** and **6**. Although not illustrated, the same description is applied to the stepped portion **72a** formed on the side wall **71a** and the reservoirs **75c** of the lid **75**. In addition, the stepped portions **72b** and **72d** formed on the side walls **71b** and **71d** have the same configuration as the stepped portion **72c** formed on the side wall **71c**.

(59) The stepped portion 72c is continuously formed all around the opening inner periphery 74c of the front surface of the side wall 71c of the frame portion 70. The stepped portion 72c includes a step inner peripheral surface 72c1, a step supporting surface 72c2, and a projection 72c3.

(60) The step inner peripheral surface 72c1 is continuously formed along the opening inner periphery 74c on the front surface side of the side wall 71c to face the housing space 72 in plan view. The step inner peripheral surface 72c1 is parallel to the Z-X plane. In addition, the height of the step inner peripheral surface 72c1 is approximately equal to the thickness (height) of the lid 75.

(61) The step supporting surface 72c2 is continuously formed along the opening inner periphery 74c of the side wall 71c to face the +Z direction and is orthogonal to the step inner peripheral surface 72c1. The width (length in the +Y direction in FIG. 6) of the step supporting surface 72c2 may be set so as to support the lid 75. The size of the step supporting surface 72c2 is set so that, when the lid 75 is attached to the stepped portion 72c, a gap G is formed between the step inner peripheral surface 72c1 and the lateral surface 75c2 of the lid 75. This gap G may be set to 0.2 mm to 0.3 mm, inclusive, for example.

(62) The projection 72c3 is formed on the step supporting surface 72c2 to extend vertically upward (in the +Z direction). The height of the projection 72c3 is set so as not to project from the front surface 75a of the lid 75. The present embodiment describes the case where the height of the projection 72c3 is so set as not to penetrate through the reservoir 75c. However, the height of the projection 72c3 may be set so that the projection 72c3 penetrates through the reservoir surface 75c1 of the lid 75 but does not project from the reservoir 75c (front surface 75a). In this case, the adhesive material 96 accumulated in the reservoir 75c adheres to the projection 72c3 penetrating through the reservoir surface 75c1 inside the reservoir 75c. This improves the bonding force of the lid 75 to the stepped portion 72c. The projection 72c3 may have a columnar shape and be circular or rectangular in plan view. The projection 72c3 may have a semispherical shape. The projection 72c3 may only be formed on the step supporting surface 72c2. The present embodiment describes, as an example, the case where the projection 72c3 is formed at a position on the step supporting surface 72c2 corresponding to the reservoir 75c formed in the lid 75.

(63) The reservoir 75c in the lid 75 is formed on the outer periphery of the front surface 75a so as to extend from the lateral surface 75c2 toward the inner side in plan view. The reservoir 75c includes a reservoir surface 75c1 that has an area positioned at a level lower than the front surface 75a at least on the side where the lateral surface 75c2 is positioned, and inner surfaces 75c3 and 75c4 formed at both sides of the reservoir surface 75c1 in the $\pm X$ directions.

(64) The reservoir surface 75c1 of the reservoir 75c of the present embodiment connects to the lateral surface 75c2 and is inclined upward from the lateral surface 75c2 toward the front surface 75a. Therefore, the reservoir surface 75c1 includes an area positioned at a level lower than the front surface 75a on the side where the lateral surface 75c2 is positioned. In this connection, as described earlier, the lateral surface 75c2 is approximately parallel to the step inner peripheral surface 72c1. The height of the lateral surface 75c2 corresponding to the reservoir 75c is approximately half the height of the lid 75. If the height of the lateral surface 75c2 corresponding to the reservoir 75c is less than this, the reservoir 75c decreases its strength and thus may be broken, which may cause a damage in the lid 75. If the height of the lateral surface 75c2 corresponding to the reservoir 75c is greater than that, the capacity of the reservoir 75c is limited to a certain amount that is insufficient for the accumulation of the adhesive material 96. If it is not possible to accumulate the adhesive material 96 sufficiently, more reservoirs 75c need to be formed or the reservoirs 75c need to be formed such that their reservoir surfaces 75c1 have a sufficient area. This may increase the cost to form the reservoirs 75c.

(65) The inner surfaces 75c3 and 75c4 connect the reservoir surface 75c1 and the front surface 75a. The inner surfaces 75c3 and 75c4 are parallel to the Z-Y plane. In this connection, each of the inner surfaces 75c3 and 75c4 and the reservoir surface 75c1 do not need to be connected orthogonally, but may be connected in a curved shape. Further, each inner surface 75c3 and 75c4 may become

toward the front surface **75a**. Still further, the reservoir surface **75c1** and the lateral surface **75c2** do not need to be connected orthogonally, but may be connected in an R-shape.

(66) In addition, the fixing hole **75b1** is formed in the rear surface **75b** of the lid **75**. The fixing hole **75b1** is formed so that, when the lid **75** is placed on the stepped portion **72c**, the fixing hole **75b1** is positioned to correspond to the projection **72c3**. The fixing hole **75b1** may only need to have a shape that allows the projection **72c3** to be inserted therein, and is formed according to the shape of the projection **72c3**. The fixing hole **75b1** may be formed to penetrate through the lid **75**, depending on the height of the projection **72c3**. The present embodiment describes, as an example, the case where the fixing hole **75b1** does not penetrate through the lid **75**.

(67) The following describes attachment of the lid **75** to the case **50** (frame portion **70**) and application of the adhesive material **96**. First, the lid **75** is attached to the stepped portions **72a**, **72b**, **72c**, and **72d** of the side walls **71a**, **71b**, **71c**, and **71d** of the frame portion **70**.

(68) Then, the adhesive material **96** is injected in the gap G between the stepped portions **72a**, **72b**, **72c**, and **72d** of the side walls **71a**, **71b**, **71c**, and **71d** and the lid **75**. In this connection, the lid **75** is fixed to the stepped portions **72a**, **72b**, **72c**, and **72d** by the projections **72a3**, **72b3**, **72c3**, and **72d3**. Therefore, the gap G from the lateral surfaces **75a2**, **75b2**, **75c2**, and **75d2** of the lid **75** is kept equal and constant.

(69) The adhesive material **96** may be injected directly in the gap G. Alternatively, the adhesive material **96** may be injected from a reservoir **75c** toward the gap G. In this case, the reservoir **75c** serves as a guide for allowing the adhesive material **96** to flow. That is, the adhesive material **96** injected in the reservoir **75c** flows through the reservoir surface **75c1** of the reservoir **75c** to the gap G. Especially, since the reservoir surface **75c1** is inclined, the adhesive material **96** flows into the gap G without fail. The adhesive material **96** flowing into the gap G in this way entirely fills the gap G. In this connection, as described earlier, the gap G all around the sides is approximately equal and kept constant. The adhesive material **96** flowing from the reservoir **75c** is easy to flow to the entire gap G stably since the gap G has no uneven part.

(70) For example, as the adhesive material **96** used here, a thermosetting resin-based adhesive agent or an organic-based adhesive agent is used. For example, the thermosetting resin-based adhesive agent contains an epoxy resin or a phenolic resin as a principal component. The organic-based adhesive agent is an elastomer-based adhesive agent containing silicone rubber or chloroprene rubber as a principal component. The adhesive material **96** preferably contains an epoxy resin or silicone rubber as a principal component. The adhesive material **96** may contain the same components as the adhesive material **60a**. However, the adhesive material **96** preferably has higher liquidity than the adhesive material **60a**. This allows the adhesive material **96** flowing from the reservoir **75c** to spread all over the gap G easily.

(71) In addition, the adhesive material **96** that fills the gap G is accumulated in the reservoirs **75c**. Therefore, it is possible to prevent the adhesive material **96** from protruding from the gap G and thus to prevent a waste of the adhesive material **96**. Since the protrusion of the adhesive material **96** is prevented, the adhesive material **96** is prevented from adhering to the lid **75** and the case **50**, which prevents defacing of the semiconductor device **10**.

(72) By solidifying the adhesive material **96** filling the gap G and being accumulated in the reservoirs **75c**, the lid **75** is firmly fixed to the stepped portions **72a**, **72b**, **72c**, and **72d** of the frame portion **70**. Since the adhesive material **96** not only fills the gap G but also flows in the reservoirs **75c**, the adhesive area between the adhesive material **96** and the lid **75** increases, as compared with the case where the reservoirs **75c** are not formed. Therefore, the bonding force of the lid **75** to the frame portion **70** (case **50**) via the adhesive material **96** is improved.

Modification Example 2-1

(73) A modification example 2-1 of the second embodiment will now be described with reference to FIG. 7. FIG. 7 is a sectional view of an area where a lid and a case are attached to each other in a semiconductor device according to the modification example 2-1 of the second embodiment. In this

connection, FIG. 7 illustrates a reservoir surface **75c1** of a reservoir **75c** that is different from that of FIG. 6, and the other structure is the same as that of FIG. 6.

(74) As illustrated in FIG. 7, in the reservoir **75c** of the modification example 2-1, the reservoir surface **75c1** connects the front surface **75a** and the lateral surface **75c2** in a curved shape. That is, in the reservoir **75c** of the modification example 2-1, the edges of the front surfaces **75a** of the lateral surface **75c2** of the lid **75** are rounded. This reservoir **75c** provides the same effects as the reservoir **75c** of FIG. 6.

(75) In this connection, a plurality of convex portions may be formed in a distributed manner on the reservoir surfaces **3c1** and **75c1** and inner surfaces **3c3**, **3c4**, **75c3**, and **75c4** of the reservoirs **3c** and **75c** of FIGS. 1, 6, and 7 as appropriate. These convex portions increase the adhesive area of the adhesive material **96** accumulated in the reservoirs **75c** to the reservoirs **75c** and improves the bonding force therebetween.

Modification Example 2-2

(76) A modification example 2-2 of the second embodiment will now be described with reference to FIG. 8. FIG. 8 is a plan view of an area where a lid and a case are attached to each other in a semiconductor device according to the modification example 2-2 of the second embodiment. In this connection, FIG. 8 illustrates inner surfaces **75c3** and **75c4** of a reservoir **75c** that are different from those of FIG. 5, and the other structure is the same as that of FIG. 5.

(77) The inner surfaces **75c3** and **75c4** of the reservoir **75c** of the modification example 2-2 are inclined in plan view. More specifically, the spacing between the inner surfaces **75c3** and **75c4** becomes wider toward the lateral surface **75c2**. In other words, the spacing between the inner surfaces **75c3** and **75c4** becomes narrower as they go from the lateral surface **75c2** in the +X direction. Since the spacing between the inner surfaces **75c3** and **75c4** becomes wider toward the gap G, the adhesive material **96** flows from the reservoir **75c** in the gap G easily. In addition, in this case, a groove may be formed along the inner surfaces **75c3** and **75c4** in the reservoir surface **75c1** to promote the adhesive material **96** to flow to the gap G. In addition, the spacing between the inner surfaces **75c4** and **75c4** may be formed to become wider in the +Z direction. This makes the adhesive material **96** to flow in the gap G more easily.

Third Embodiment

(78) In a third embodiment, a semiconductor device different from that of the second embodiment will be described with reference to FIGS. 9 to 11. FIG. 9 is a plan view of the semiconductor device according to the third embodiment. FIG. 10 is a plan view of the inside of the semiconductor device according to the third embodiment. FIG. 11 is a sectional view of the semiconductor device according to the third embodiment. In this connection, FIG. 11 is a sectional view taken along the dash-dotted line X-X of FIG. 9. The same reference numerals as used in the second embodiment are given to the corresponding components and the description of them may be omitted or simplified.

(79) The semiconductor device **10a** of the third embodiment includes an insulated circuit substrate **20a**, a case **50a** for accommodating the insulated circuit substrate **20a**, and a lid **75**. The inside of the case **50a** is sealed with a sealing material **95**. In addition, a metal base substrate **23** is exposed downward (in the -Z direction) from the rear surface of the case **50a** of the semiconductor device **10a**.

(80) The case **50a** is rectangular in plan view. The case **50a** includes a frame portion **70a** and fastening holes **73a**, **73b**, **73c**, and **73d**. The frame portion **70a** is rectangular (frame-shaped) in plan view. The frame portion **70a** has a housing space **72** at the center thereof, and has side walls **71a**, **71b**, **71c**, and **71d** in order at the four sides of the housing space **72**. More specifically, the side walls **71a** and **71c** are arranged at the long sides of the frame portion **70a**, and the side walls **71b** and **71d** are arranged at the short sides thereof. In this connection, the frame portion **70a** may only have flat side walls **71a**, **71b**, **71c**, and **71d** at the four sides of the rectangle and may have a curve (curved surface) at each corner.

(81) As in the second embodiment, the side walls **71a**, **71b**, **71c**, and **71d** have stepped portions **72a**, **72b**, **72c**, and **72d**. Projections **72a3**, **72b3**, **72c3**, and **72d3** are formed at the stepped portions **72a**, **72b**, **72c**, and **72d**.

(82) In addition, a plurality of terminals extend vertically upward (in the +Z direction) from the front surface of the frame portion **70a** along the front surface of the frame portion **70a**. The plurality of terminals include external connection terminals for the main electrodes, control, and output of an insulated circuit substrate (inverter), which will be described later. The external connection terminals for main electrode include second P terminals **184a**, second N terminals **185a**, first P terminals **184b**, and first N terminals **185b**. The second P terminals **184a** and second N terminals **185a** are formed on the side wall **71d** so as to be aligned in the direction from the side wall **71c** toward the side wall **71a**. The first P terminals **184b** and first N terminals **185b** are formed on the side wall **71b** so as to be aligned in the direction from the side wall **71c** toward the side wall **71a**.

(83) The external connection terminals for control include a Gx terminal **191**, a Gu terminal **192**, a Gy terminal **193**, a Gv terminal **194**, a Gz terminal **195**, and a Gw terminal **196**, which are formed on the side wall **71a** so as to be aligned in the direction from the side wall **71b** toward the side wall **71d**. The Gx terminal **191**, Gu terminal **192**, Gy terminal **193**, Gv terminal **194**, Gz terminal **195**, and Gw terminal **196** each have one end extending upward from the top surface of the side wall **71a**. In addition, the Gx terminal **191**, Gu terminal **192**, Gy terminal **193**, Gv terminal **194**, Gz terminal **195**, and Gw terminal **196** each have the other end exposed on the terminal stepped portion **71e**.

(84) The external connection terminals for output include U terminals **181**, V terminals **182**, and W terminals **183**, which are formed on the side wall **71c** so as to be aligned in the direction from the side wall **71b** toward the side wall **71d**. The U terminals **181**, V terminals **182**, and W terminals **183** each have one end extending upward from the top surface of the side wall **71c**. In addition, the U terminals **181**, V terminals **182**, and W terminals **183** each have the other end exposed on the terminal stepped portion **71f**.

(85) The plurality of terminals are made of a material with high electrical conductivity, such as copper, aluminum, iron, or an alloy containing at least one of these. In addition, plating may be performed on the surfaces of the plurality of terminals using, for example, nickel, gold, tin, and an alloy containing at least one of these, to improve their corrosion resistance.

(86) The semiconductor device **10a** configured as above includes an inverter circuit. The inverter circuit is designed to convert direct current to alternating current. Direct current input from the second P terminals **184a** and first P terminals **184b** with high potential and the second N terminals **185a** and first N terminals **185b** with low potential is converted by a plurality of semiconductor chips **31a** and **32a** to three-phase alternating current, which is then output from the U terminals **181**, V terminals **182**, and W terminals **183**.

(87) The lid **75** is provided so as to cover the opening of the housing space **72** for accommodating the components (semiconductor chips and others) of the semiconductor device **10a** (see FIG. 9). The fastening holes **73a**, **73b**, **73c**, and **73d** are provided at the four corners of the frame portion **70a** in plan view. Screws are inserted into the fastening holes **73a**, **73b**, **73c**, and **73d**. The semiconductor device **10a** is attached to a predetermined position by inserting the screws into the fastening holes **73a**, **73b**, **73c** and **73d**. These screws pass through the fastening holes **73a**, **73b**, **73c**, and **73d** and also, for example, into the through holes (which will be described later) of a cooling unit, so that the cooling unit is screwed together with the case **50a**.

(88) This case **50a** is made of the same resin as used in the second embodiment and is formed by integral molding with the plurality of terminals in the same manner as in the second embodiment. The lid **75** is formed separately, using the same material as the case **50a**. The same metal base substrate **23** as used in the second embodiment is used here. Such a metal base substrate **23** is firmly fixed to the rear opening **63** of the rear surface of the frame portion **70** via an adhesive

material.

(89) In addition, in the semiconductor device **10a**, three insulated circuit substrates **20a** are disposed along the side walls **71a** and **71c** in the housing space **72** of the case **50a** on the metal base substrate **23**. Each insulated circuit substrate **20a** includes an insulating plate **21**, circuit patterns **22** formed on the insulating plate **21**, and a metal plate **23a** formed on the rear surface of the insulating plate **21**. The insulating plate **21** and circuit patterns **22** are made of the same materials as used in the second embodiment. Note that the shapes and quantity of the circuit patterns **22** are different from those provided in the second embodiment.

(90) The metal plate **23a** is formed on the entire rear surface of the insulating plate **21** except the outer periphery thereof. The metal plate **23a** is made of a metal with high thermal conductivity as a principal component. Examples of the metal include copper, aluminum, and an alloy containing at least one of these. The thickness of the metal plate **23a** is in the range of 0.1 mm to 5.0 mm, inclusive. Plating may be performed on the surface of the metal plate **23a** to improve its corrosion resistance. Examples of the plating material used here include nickel, a nickel-phosphorus alloy, and a nickel-boron alloy.

(91) As each insulated circuit substrate **20a** configured as above, a DCB substrate or AMB substrate may be used, for example. The insulated circuit substrate **20a** is able to transfer heat generated by the semiconductor chips **31a** and **32a** through the circuit patterns **22**, insulating plate **21**, and metal plate **23a** to the outside.

(92) The semiconductor chips **31a** and **32a** are made of silicon, silicon carbide, or gallium nitride as a principal component. The semiconductor chips **31a** and **32a** include reverse-conducting (RC)-IGBT chips. The semiconductor chips **31a** and **32a** of this type each have an input electrode (collector electrode) serving as a main electrode on the rear surface thereof and have a gate electrode serving as a control electrode and an output electrode (emitter electrode) serving as a main electrode on the front surface thereof.

(93) The above insulated circuit substrates **20a** and the semiconductor chips **31a** and **32a** are accommodated, and the circuit patterns **22**, semiconductor chips **31a** and **32a**, and the plurality of terminals are electrically and mechanically connected as appropriate with wires (reference numerals not given). As a result, the inverter circuit is provided in the housing space **72**.

(94) As in the second embodiment, reservoirs **75c** are formed in the lid **75** that covers the insulated circuit substrates **20a** and semiconductor chips **31a** and **32a** accommodated in the frame portion **70a** of the case **50a**. In the third embodiment, five reservoirs **75c** are formed at each of the opposite long sides of the front surface **75a** of the lid **75**. Note that, as described earlier, the reservoirs **75c** are formed, taking into consideration the warpage of the case **50a** caused by screwing through the fastening holes **73a**, **73b**, **73c**, and **73d**.

(95) In addition, some of the reservoirs **75c** are formed at positions corresponding to the centers of the side walls **71a** and **71c** of the frame portion **70a**, in the lid **75**. When the semiconductor device **10a** repeatedly experiences temperature changes during operation, it is warped due to a difference in linear expansion coefficient among its components. Since the semiconductor device **10a** is rectangular in plan view, it is warped in a concave or convex shape with its center as a vertex in plan view. At this time, stress is generated at the centers of the side walls **71a** and **71c** and the centers of the side walls **71b** and **71d**. If no reservoir **75c** is formed in the lid **75**, the lid **75**, even bonded to the frame portion **70a** using the adhesive material **96**, may be separated from the frame portion **70a** due to such stress. By contrast, by forming the reservoirs **75c** at the positions where the stress is to be generated, the reservoirs **75c** improve the bonding force of the adhesive material **96** and thus prevent the separation of the lid **75** from the frame portion **70a**.

(96) The third embodiment describes the case where no reservoir **75c** is formed on the sides of the lid **75** where the side walls **71b** and **71d** are positioned. In the case where reservoirs **75c** are formed on the sides of the lid **75** where the side walls **71b** and **71d** are positioned, the reservoirs **75c** are preferably formed at positions corresponding to the centers of the side walls **71b** and **71d**.

(97) In addition, the reservoirs 75c of the third embodiment may be formed such that their reservoir surfaces are parallel to the front surface 75a, as with the reservoirs 3c of the first embodiment. In addition, the reservoirs 75c of the third embodiment may be formed such that their reservoir surfaces are curved as illustrated in FIG. 7 or the spacing between their opposite inner surfaces gradually become wider as illustrated in FIG. 8.

(98) The disclosed techniques make it possible to prevent an increase in the size of a semiconductor device and to prevent protrusion of an adhesive material.

(99) All examples and conditional language provided herein are intended for the pedagogical purposes of aiding the reader in understanding the invention and the concepts contributed by the inventor to further the art, and are not to be construed as limitations to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although one or more embodiments of the present invention have been described in detail, it should be understood that various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Claims

1. A semiconductor device, comprising: a semiconductor element; a case including a frame portion, which is of a frame shape having an opening in a plan view of the semiconductor device, the frame portion having a front surface, and an inner periphery that surrounds a housing space for accommodating the semiconductor element; and a lid having a flat plate shape, the lid covering an upper side of the housing space, wherein the inner periphery of the frame portion has a stepped portion formed thereon, the stepped portion including a step supporting surface positioned at a level lower than the front surface of the frame portion and being approximately parallel to the front surface of the frame portion, the lid has a front surface and a bottom surface, and a lateral surface surrounding the lid, the lateral surface being bonded to the stepped portion via an adhesive material, such that the front surface and the bottom surface of the lid are approximately parallel to the step supporting surface, and the lid has a reservoir formed in the front surface thereof and extending from the lateral surface, the reservoir having a reservoir surface substantially lower than the front surface of the lid.

2. The semiconductor device according to claim 1, wherein the reservoir surface is inclined upward from the lateral surface of the lid toward the front surface of the lid in a sectional view of the semiconductor device.

3. The semiconductor device according to claim 1, wherein the reservoir surface is curved in a sectional view of the semiconductor device.

4. The semiconductor device according to claim 1, wherein the reservoir surface is approximately parallel to the front surface of the lid in a sectional view of the semiconductor device.

5. The semiconductor device according to claim 1, wherein the stepped portion is continuously formed along the entire inner periphery of the frame portion.

6. The semiconductor device according to claim 5, wherein the frame portion has a projection formed on the step supporting surface of the stepped portion, and the bottom surface of the lid at an outer periphery thereof engages with the projection so that the lid is supported by the step supporting surface.

7. The semiconductor device according to claim 1, wherein the stepped portion further includes an inner peripheral surface connected to the step supporting surface and being approximately perpendicular to the step supporting surface, and the lid is disposed such that the lateral surface of the lid is approximately parallel to the inner peripheral surface of the stepped portion, with a gap therebetween.

8. The semiconductor device according to claim 7, wherein the gap is filled with the adhesive

material.

9. The semiconductor device according to claim 1, wherein the reservoir surface of the reservoir in the lid bonded to the stepped portion connects to the lateral surface, and the lateral surface is approximately parallel to an inner peripheral surface of the stepped portion.
 10. The semiconductor device according to claim 1, wherein a front surface of the adhesive material is positioned at a level lower than the front surface of the frame portion.
 11. The semiconductor device according to claim 1, wherein the reservoir is formed at an outer periphery of the lid.
 12. The semiconductor device according to claim 11, wherein the front surface of the lid has two sides opposite to each other, and the reservoir is provided in two, each formed at least at a center of one of the two sides in the plan view.
 13. The semiconductor device according to claim 11, wherein the frame portion of the case has two sides opposite to each other, the case further includes two fastening holes respectively at the two opposite sides in the plan view, the fastening hole being open in a direction substantially perpendicular to the front surface of the frame portion, and the reservoir is provided in two, each formed at a position corresponding to one of the fastening holes.
 14. The semiconductor device according to claim 1, wherein the reservoir surface has a rectangular shape in the plan view.
 15. The semiconductor device according to claim 1, wherein in the plan view, the reservoir surface has a trapezoid shape, including a first side and a second side that is parallel to the first side and shorter than the first side, the first side overlapping the lateral surface of the lid.
 16. The semiconductor device according to claim 1, wherein the lid further includes a positioning member at a predetermined position on the front surface of the lid and extend vertically upward, and the reservoir is formed at an area other than a vicinity of the predetermined position.
 17. The semiconductor device according to claim 1, wherein the reservoir is continuously formed along an entire outer periphery of the lid.
 18. The semiconductor device according to claim 1, wherein the case further includes an external connection terminal integrally formed with the frame portion, the external connection terminal having two ends, one end thereof extending vertically upward from the front surface of the frame portion and the other end thereof extending into the housing space.
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